

OPERATION: REBUILD FROM CHAOS

Official Model Reconstruction Challenge Operations Manual

OPERATIVE CHALLENGE BRIEF

This document governs the official NSOC model recovery parameters. Operatives are instructed to proceed with analytical exploration and reassemble the corrupted neural configurations locally.

1. EVENT OVERVIEW

The Model Reconstruction Challenge is an elite, high-intensity machine learning and reverse-engineering competition. Operatives are dropped into a digital scenario where a trained deep residual neural network has been shattered into unlabelled weight matrices and bias tensors. Your mission is to apply mathematical forensics to reconstruct the network and recover the core classifications.

This operations exercise has been designed by NSOC to evaluate:

- **Deep Neural Network Topologies:** Structural mapping of parameters and connections.
- **Reverse-Engineering Diagnostics:** Analytical heuristics of trained weight variables.
- **Combinatorial Optimization:** Search algorithms, matching strategies, and graph reassembly.

2. FORENSIC MISSION & OBJECTIVES

A trained residual network model classification head has been compromised. All N constituent layer tensors have been separated into a chaotic collection (piece_0.pth to piece_N.pth). Your objective is to:

- 1. Identify Component Types:** Segregate the layers based on their weight tensor shapes.
- 2. Match and Pair Components:** Correctly pair input projections with their corresponding output projections.
- 3. Sequence Residual Blocks:** Discover the precise sequential order of the paired blocks from 1 to K.
- 4. Restore Grid Logits:** Reassemble the original architecture to match predictions and recover classifications.

3. THE THREE PHASES OF ALIGNMENT

Phase 1: Distribution

Operatives receive the workspace package containing the scrambled layers, calibration data, and baseline code.

Phase 2: Reconstruction

Operatives run exploratory analysis, develop pairing strategies, map the block sequences, and locally optimize their models.

Phase 3: Submission

Teams package their automated pipelines, reconstructed weights, submission coordinates, and technical report.

4. COMPILER REPOSITORY STRUCTURE & CLONE GUIDE

To streamline deployment and code syncing, all assets are tracked under a local workspace repository. Operatives should initialize their local grids as follows:

```
# Clone the central NSOC challenge repository
git clone https://github.com/nsoc-club/model-reconstruction-chaos.git
cd model-reconstruction-chaos

# Spawn your dedicated local Team branch (Do not push to main!)
git checkout -b team-[your_team_name]

# Install required dependencies
pip install -r requirements.txt
```

Operational File Topology

The repository contains the following files:

- data/pieces/: Scrambled weight fragments (piece_0.pth to piece_N.pth)
- data/historical_data.csv: Calibration alignment telemetry
- data/metadata.json: Cryptographic hashes for validation
- samples/sample_submission.csv: Standard submission configuration layout
- starter_kit.ipynb: Guided interactive forensic notebook
- verify_submission.py: Local score validation utility

5. SECURED TELEMETRY (DATASET ANALYSIS)

The provided calibration dataset serves as intermediate alignment coordinates to verify your reassembly pipeline. The primary file is data/historical_data.csv. It represents representative input samples processed by the original model, capturing input feature dimensions (pixel_0 to pixel_X), predicted target logits (pred_logit_0 to pred_logit_Y), and target classes.

6. DISMANTLED NEURAL CORE (MODEL COMPONENTS)

The N scrambled files must be classified strictly based on their weight tensor shapes:

Component	Qty	Shape	Operational Role
Front Projection (proj)	1	(D_latent, D_in)	Maps input space dimensions down to latent space features.
Classification Head (last)	1	(D_classes, D_latent)	Classifies latent features into output classes.
Input Projections (W_in)	K	(D_sub, D_latent)	Projects intermediate features up to a sub-block space.
Output Projections (W_out)	K	(D_latent, D_sub)	Projects sub-block features back down to latent space.

Residual Block Equation

Each block k projects latent features x using the residual mapping:

Block_k(x) = x + W_out @ ReLU(W_in @ x + b_in) + b_out

7. ALLOWED FILE FORMATS

Only standard code and configuration extensions are allowed during submission:

- Approved: .py, .ipynb, .csv, .pth, .txt, .json
- Prohibited: Compiled executable binaries or external model weights.

8. TEAM RULES & 9. ARSENAL POLICY

- **Team Size:** 1 to 3 registered operatives.
- **Independence:** Complete isolation is required. Inter-team collusion will lead to disqualification.
- **Allowed Tools:** Standard Python libraries (PyTorch, SciPy, Scikit-Learn), manuals, and papers.
- **Prohibited Tools:** Web-based black-box solving APIs or sharing private logic blocks.

10. SUBMISSION & 11. AFTER-ACTION REPORT

Teams must submit a compressed package named [team_name].zip containing:

- solution.py: Fully automated reassembly script
- final_model.pth: Reconstructed model weights
- submission.csv: Reassembled indices spreadsheet
- report.pdf: Exhaustive technical report

Report Core Outlines

Your Technical Report must cover: Introduction, Methodology, Implementation details, Experiments/Results, Challenges Faced, and Key Learnings.

12. SCORE GRID & EVALUATION CRITERIA

Scores are calculated as: $\text{Final Score} = 0.70 * \text{Model Score} + 0.30 * \text{Report Score}$

Evaluation Area	Marks	Target Metric / Criteria
Model Logits MSE	50 Marks	MSE on intermediate classification logits
Reconstruction Quality	10 Marks	Correct layer pairing and sequential reassembly
Runtime & Efficiency	10 Marks	Speed and computational footprint of the solver
Technical Explanation	10 Marks	Mathematical depth and clarity in the report
Methodology & Innovation	8 Marks	Rigor and uniqueness of the pairing and search strategies
Experimental Analysis	6 Marks	Quality of testing, parameter logs, and performance curves
Presentation & Clarity	4 Marks	Document structure and readability
Reproducibility	2 Marks	Cleanliness of configuration guides and scripts

13. VERIFICATION & JUDGING DISCRETION

JUDGING PANEL RULE

The final evaluation, grading decisions, score validation, and official rankings rest solely at the absolute discretion of the NSOC organizing team. The decisions of the judging panel are final and absolute.

Submissions utilizing hardcoded indices, memorized mappings, or bypassing the core reassembly logic will fail verification and receive a score of 0.0.

14. STABILIZATION TIE-BREAKERS

Submissions are evaluated against an isolated hidden dataset to prevent overfitting. If scores are equal, the following automated tie-breaking hierarchy applies:

1. Lowest Logits MSE -> 2. Highest Pairing Accuracy -> 3. Exact Sequence Match -> 4. Least Forward Evaluations -> 5. Earliest Submission Timestamp.

15. RED LINES & 16. NSOC PHILOSOPHY

Violations such as plagiarism, coordinate sharing, dataset tampering, or proxy registrations will lead to immediate disqualification. The NSOC Command prizes mathematical elegance, creative problem solving, and computational efficiency.